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1. Introduction

1.1 Background

Access to quality education remains one of the most pressing challenges facing developing nations, particularly in Sub-Saharan Africa. In Sierra Leone, structural barriers including inadequate school infrastructure, a shortage of qualified teachers, and limited access to learning materials continue to prevent millions of students from receiving the education they deserve. These challenges are further compounded in rural and remote communities where resources are scarce and connectivity is limited.

EduBridge is a proposed Digital Public Good (DPG) designed to address these barriers through technology. By leveraging the widespread availability of mobile devices and internet connectivity, EduBridge provides a free, accessible, and scalable platform that connects students with structured learning content, volunteer mentors, interactive quizzes, and peer learning communities.

1.2 Problem Statement

Millions of students in underserved communities — including rural areas, low-income neighbourhoods, and displaced populations — lack access to structured learning materials, qualified mentors, and academic guidance. This persistent education gap perpetuates cycles of poverty and inequality, leaving entire generations without the skills needed to participate in the modern economy.

1.3 Project Objectives

- Apply Software Development Life Cycle (SDLC) concepts to plan and document the EduBridge system.
- Design a complete and scalable system architecture using a 3-tier client-server model.
- Develop and interpret UML diagrams including Use Case, Class, Sequence, and Activity diagrams.
- Create a functional 12-screen high-fidelity prototype using Figma, designed responsively for both mobile and desktop.

-
- Establish a public GitHub repository with proper version control, a documented backend structure, and collaboration history.
 - Produce a comprehensive technical report aligned with the Digital Public Goods standard.

1.4 SDG 4 Alignment

EduBridge directly addresses United Nations Sustainable Development Goal 4: Quality Education — specifically the mandate to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. The platform targets three key sub-targets: Target 4.1 (free primary and secondary education), Target 4.3 (equal access to technical and vocational education), and Target 4.5 (elimination of gender disparities and barriers for vulnerable populations).

1.5 Scope

The scope of this project covers the complete software design of the EduBridge platform, including the SDLC methodology, system architecture, UML diagrams, a responsive Figma prototype, project management artefacts, GitHub integration with a documented backend structure, and technical documentation. The project does not include the full implementation or live deployment of the system; rather, it produces a comprehensive, buildable design blueprint.

2. Software Development Life Cycle (SDLC)

2.1 Selected Model: Agile (Scrum-Based Iterative Development)

EduBridge will be developed using the Agile Software Development Life Cycle model, specifically the Scrum framework. Agile was selected because it allows the development team to build the system in small, functional increments called sprints, enabling continuous feedback from students, mentors, and educators throughout the process. This is especially important for a Digital Public Good, where user needs evolve rapidly and the platform must remain responsive to community requirements.

Unlike the Waterfall model, which requires all requirements to be defined upfront, Agile accommodates change at every stage. This flexibility is critical for EduBridge, as new educational content, features such as gamification, and language support will need to be added iteratively based on real user data.

2.2 Phase-by-Phase Explanation

Phase 1 — Planning

The Planning phase defines the project scope before development begins. For EduBridge, this involves identifying the core problem — education inequality in underserved communities — and defining the platform's scope as a web-based application offering free courses, mentor matching, progress tracking, interactive quizzes, and a community forum.

Key stakeholders are identified: primary and secondary school students, volunteer mentors and tutors, school administrators, and the Ministry of Basic and Senior Secondary Education (MBSSE) in Sierra Leone. Tools are selected: Figma for UI/UX prototyping, GitHub for version control, draw.io for diagrams, and Node.js with PostgreSQL for the technology stack. A project timeline of 11 weeks is established using a Gantt chart, and the development budget is confirmed as zero-cost using open-source tools.

Deliverable: Project charter, stakeholder register, project timeline, tool selection documentation.

Phase 2 — Requirements Analysis

Functional and non-functional requirements are gathered through interviews with potential users, analysis of existing educational platforms, and review of the Sierra Leonean national curriculum. The output is a Software Requirements Specification (SRS) document.

ID	Functional Requirement	Priority
FR-01	Student registration and login	High
FR-02	Browse and enroll in courses by subject and grade level	High
FR-03	Access learning materials including notes, videos, and quizzes	High
FR-04	Track personal learning progress and quiz scores	High
FR-05	Request a volunteer mentor by subject	High
FR-06	Mentor accepts or rejects mentorship requests	Medium
FR-07	Student-mentor messaging and chat	Medium
FR-08	Take practice quizzes and view scored results	High

FR-09	Post and reply in community forum	Medium
FR-10	Admin manages users, courses, and content	High

ID	Non-Functional Requirement
NFR-01	Must function on low-bandwidth (3G) internet connections
NFR-02	Responsive design supporting both mobile and desktop browsers
NFR-03	All user data encrypted using HTTPS and JWT authentication
NFR-04	Accessible to users with disabilities in compliance with WCAG 2.1
NFR-05	Page load time must not exceed 3 seconds under normal conditions

Deliverable: Software Requirements Specification (SRS) document.

Phase 3 — System Design

In the Design phase, the technical blueprint of EduBridge is created. The system architecture is defined as a 3-tier client-server model. UML diagrams are produced including the Use Case Diagram, Class Diagram, Sequence Diagram, and Activity Diagram. The database schema is designed with nine tables. The UI/UX is prototyped in Figma with 12 key screens, designed in both mobile and desktop responsive formats to cover all primary user journeys.

Deliverable: Architecture document, UML diagrams, database schema, responsive Figma prototype.

Phase 4 — Implementation

Development is divided into three Agile sprints. Sprint 1 covers user authentication and profile setup. Sprint 2 covers the course module, learning materials, and the Q&A forum. Sprint 3 covers the quiz engine, progress tracking, and the mentor matching system. The frontend is built with React.js, the backend with Node.js and Express, and the database with PostgreSQL. All source code is managed through Git with pull request-based code reviews.

Deliverable: Working backend source code structure hosted in the public GitHub repository.

Phase 5 — Testing

Each sprint concludes with a testing cycle. Unit testing validates individual functions such as login validation and quiz auto-grading. Integration testing confirms that modules communicate correctly, for example that a course enrolment triggers a progress record update. User Acceptance Testing (UAT) is conducted with a group of 10 students and 3 mentors. Bugs are logged using GitHub Issues and resolved before the next sprint begins.

Deliverable: Test reports, bug logs, refined prototype.

Phase 6 — Deployment

The EduBridge web application is deployed on a free cloud hosting platform such as Vercel or Heroku. Database hosting is configured on a cloud provider. The GitHub repository is made public with a comprehensive README and a documented backend folder structure. The MIT open-source licence is applied, and the project is prepared for submission to the Digital Public Goods Alliance (DPGA) for official recognition.

Deliverable: Live application URL, public repository, deployment documentation.

Phase 7 — Maintenance

Following deployment, the team monitors the system for bugs, performance degradation, and security vulnerabilities. New course content is added regularly based on the national curriculum. User feedback collected through an in-app form drives the

next development sprint. Future planned features include gamification badges, multilingual support in Krio and Temne, and an offline mode for low-connectivity areas.

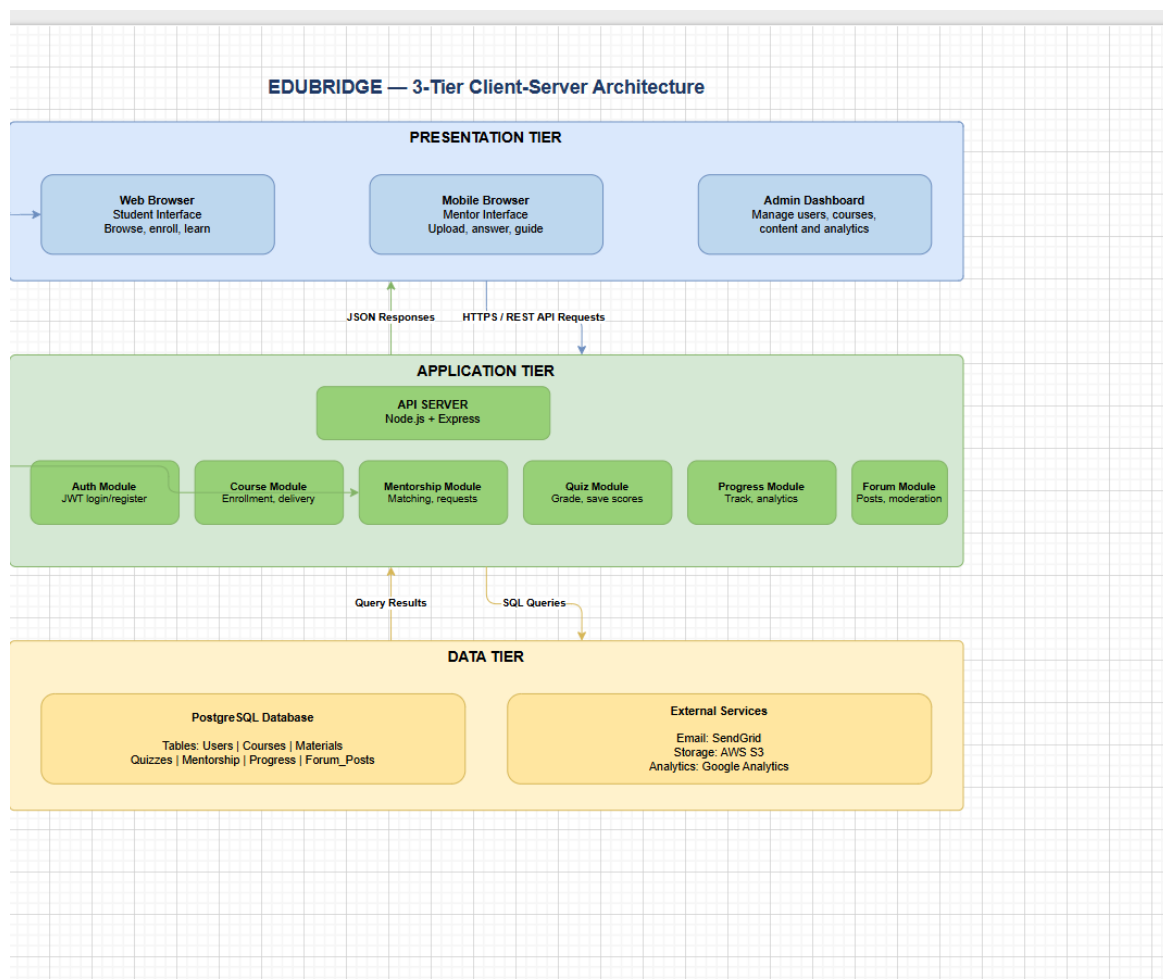
Deliverable: Changelog, update releases, community contribution guidelines.

3. System Architecture Design

3.1 Architecture Overview

EduBridge is built on a three-tier client-server architecture. This architectural style was chosen because it separates concerns across three distinct layers — the user interface, the business logic, and the data storage — making the system significantly easier to maintain, scale, and secure independently. Each tier can be updated or replaced without affecting the others, which is essential for a growing Digital Public Good platform.

3.2 Architecture Diagram



3.3 Component Descriptions

Presentation Layer (Tier 1)

This is the topmost tier and represents everything the user sees and interacts with. EduBridge provides three distinct interfaces within this layer: a Student Interface for browsing courses, accessing materials, taking quizzes, and tracking progress; a Mentor Interface for accepting student requests, uploading resources, and communicating with mentees; and an Admin Dashboard for managing users, courses, and platform content. All three interfaces are built as responsive web applications accessible through any modern browser, optimised for both desktop and low-bandwidth mobile connections.

Application Layer (Tier 2)

This is the middle tier and serves as the processing engine of EduBridge. It is built using Node.js with the Express framework and exposes a RESTful API that the frontend communicates with exclusively. The EduBridge backend contains six core route modules, each handling a distinct area of business logic:

- **Auth Module:** Handles user registration, login, and session management using JWT (JSON Web Token) authentication.
- **Course Module:** Manages course listings, student enrolment, and lesson content delivery.
- **Mentorship Module:** Matches students with available mentors based on subject area and availability, and manages request status.
- **Quiz Module:** Delivers timed multiple-choice quizzes, auto-grades student responses, and stores results.
- **Progress Module:** Tracks each student's completed lessons, quiz scores, and generates analytics.
- **Forum Module:** Manages community discussion posts, replies, and content moderation.

All API endpoints are secured using JWT middleware that validates each incoming request before processing begins. The complete backend folder structure, including

the server configuration, all six route files, and the database schema, is available in the /backend folder of the public GitHub repository (see Section 7).

Data Layer (Tier 3)

This is the bottommost tier and handles all persistent data storage. EduBridge uses a PostgreSQL relational database containing nine tables: Users, Courses, Enrollments, Materials, Quizzes, Quiz_Results, Progress, Mentorship, and Forum_Posts. Tables are linked through foreign key relationships to maintain data integrity — for example, the Enrollments table references both the Users table and the Courses table, establishing the relationship between a student and their enrolled courses. All database queries from the Application Layer use parameterised SQL statements to prevent SQL injection attacks and ensure security. The complete schema, including sample data, is documented in backend/database/schema.sql within the GitHub repository.

EduBridge also integrates with three external services: an Email Service (SendGrid) for sending registration confirmations and notifications; Cloud Storage (AWS S3 or Firebase Storage) for hosting uploaded videos and documents; and an Analytics Service (Google Analytics) for tracking platform usage and learning engagement patterns.

3.4 Component Interaction and Data Flow

When a student performs an action on EduBridge — for example, enrolling in a course — the following data flow occurs. The student clicks the Enroll button on the Course Detail screen. The Presentation Layer sends an HTTPS POST request containing the student's JWT token and the course ID to the Application Layer's Course Module endpoint (POST /api/courses/:id/enroll). The Application Layer first verifies the JWT token through the Auth Module to confirm the user is authenticated and authorised. It then executes a SQL INSERT query on the Enrollments table in the PostgreSQL database to record the enrolment. The database confirms the operation and returns a success response. The Application Layer formats this as a JSON response and sends it back to the frontend. The Presentation Layer then updates the UI to show the course as enrolled with a progress bar initialised at 0%.

3.5 Technology Stack

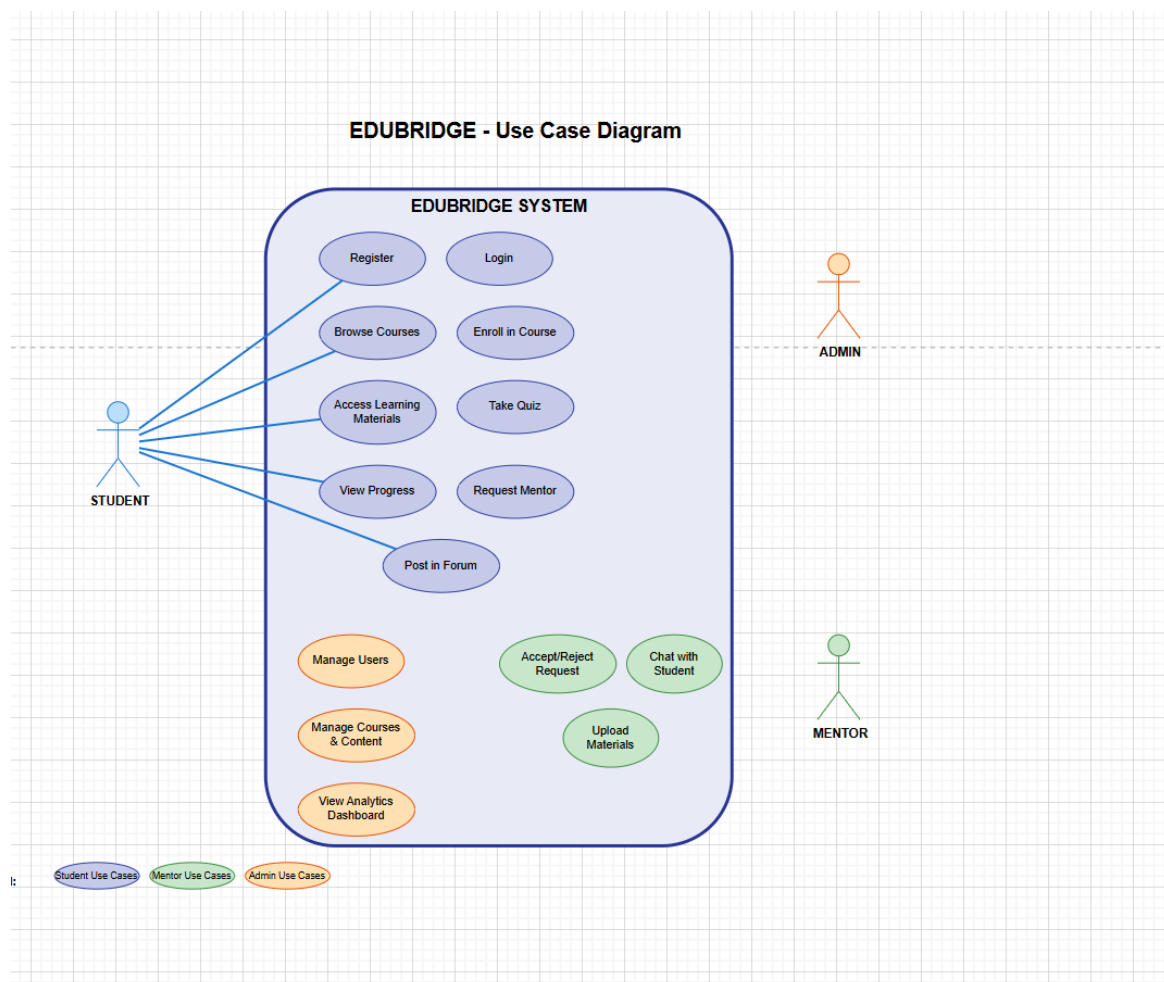
Layer	Technology / Tool
Frontend	React.js — responsive for mobile and desktop
Backend	Node.js + Express.js (REST API)
Database	PostgreSQL (Relational Database)
Authentication	JWT — JSON Web Token
File Storage	AWS S3 / Firebase Storage
Email Service	SendGrid
Version Control	Git + GitHub
Prototype Tool	Figma (Mobile + Desktop)
Deployment	Vercel / Heroku (free tier)
Diagram Tool	draw.io (diagrams.net)

4. UML Diagrams

This section presents four UML diagrams that model the structure and behaviour of the EduBridge system, each providing a different perspective of the system design.

4.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the three actors — Student, Mentor, and Admin — and the functions available to each within the EduBridge system, including registration, course enrollment, quiz taking, mentor matching, and forum participation. Admin use cases connect to Manage Users, Manage Courses & Content, and View Analytics Dashboard, and Login connects to all three actors.



4.2 Class Diagram

The Class Diagram presents the static structure of the EduBridge system, showing the six core classes, their attributes, methods, and the relationships between them: User, Material, Question, Answer, Progress, and LiteracyProgram, interconnected through associations, dependencies, and aggregation relationships.

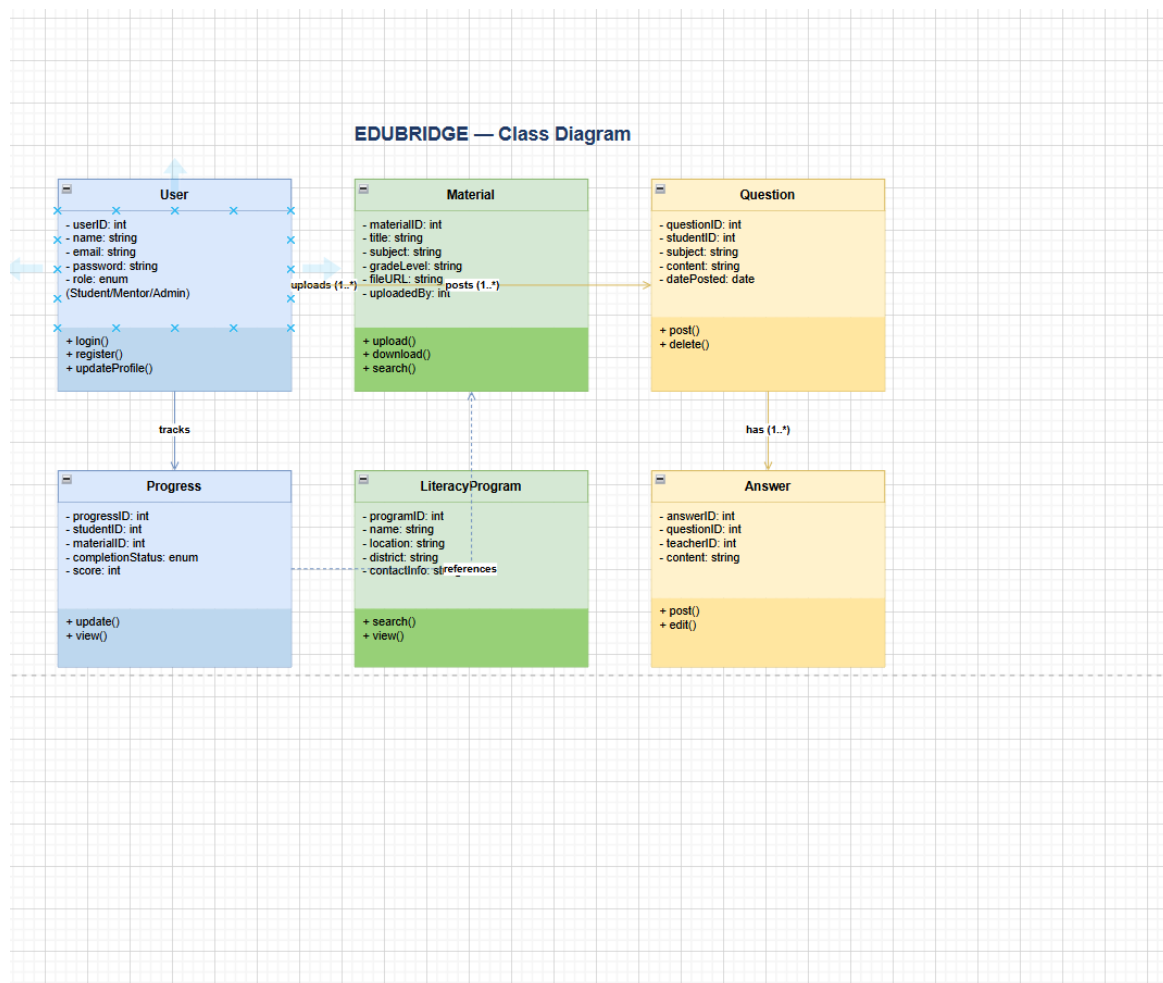


Figure 3: EduBridge Class Diagram

4.3 Sequence Diagram — Student Takes a Quiz

The Sequence Diagram models the step-by-step interaction between the Student, the Browser, the Server, and the Database during the quiz-taking process. The thirteen-step sequence begins when the student clicks Start Quiz, triggering a GET request to `/api/quiz/123`. The server queries the database with `SELECT * FROM quizzes WHERE id=123`, and the quiz data is returned and displayed to the student. After the student submits answers via `POST /api/quiz/submit`, the server auto-grades the responses,

saves the result with an INSERT INTO quiz_results query, and returns the final score and percentage to be displayed on the results page.

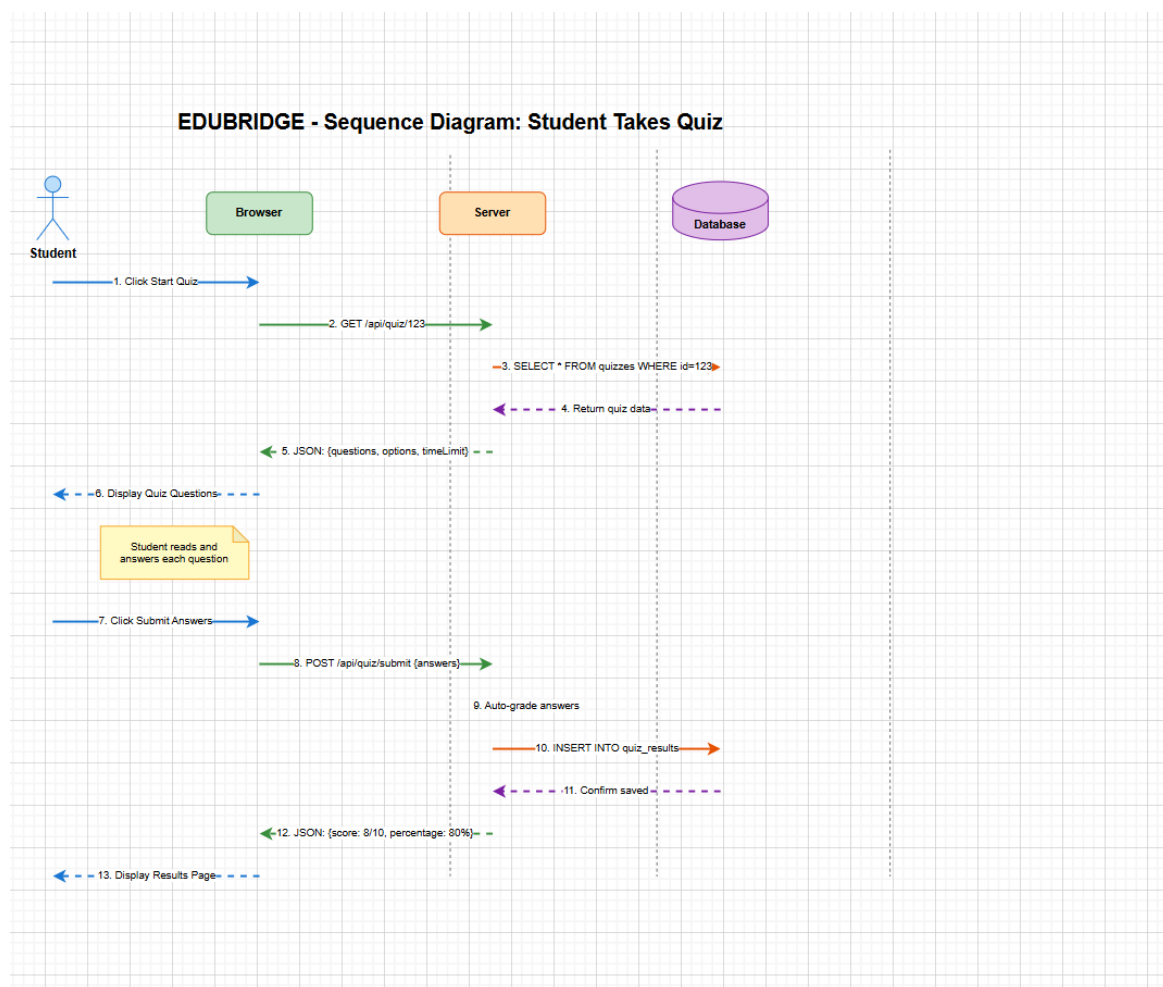


Figure 4: EduBridge Sequence Diagram — Student Takes a Quiz

4.4 Activity Diagram — User Registration Flow

The Activity Diagram models the complete user registration and login process in EduBridge. It shows the parallel decision paths available to new users (registration) and returning users (login), including validation steps, role selection between Student and Mentor, and the final routing to the Student Dashboard upon successful authentication.

EDUBRIDGE - Activity Diagram: User Registration Flow



5. Prototype Development

5.1 UI/UX Design Principles

The EduBridge user interface was designed according to three guiding principles: simplicity, consistency, and accessibility. Every design decision prioritises the experience of a first-time user in Sierra Leone, whether accessing the platform on a budget Android smartphone or a shared desktop computer in a school computer lab.

The colour palette uses Primary Blue (#2563EB) as the dominant brand colour, chosen for its association with trust, technology, and academic excellence. Secondary Green (#10B981) is used for success states and positive feedback. The background remains white (#FFFFFF) with dark text (#1F2937) to maximise readability and contrast compliance with WCAG 2.1 AA standards.

Typography uses the Inter typeface throughout for its excellent legibility at small sizes across all device types. Navigation follows a bottom navigation bar pattern on mobile, placing the five core sections — Home, Courses, Mentor, Forum, and Profile — within easy thumb reach. All interactive elements maintain a minimum tap target size of 48px in height to comply with mobile usability standards.

5.2 Responsive Design — Mobile and Desktop

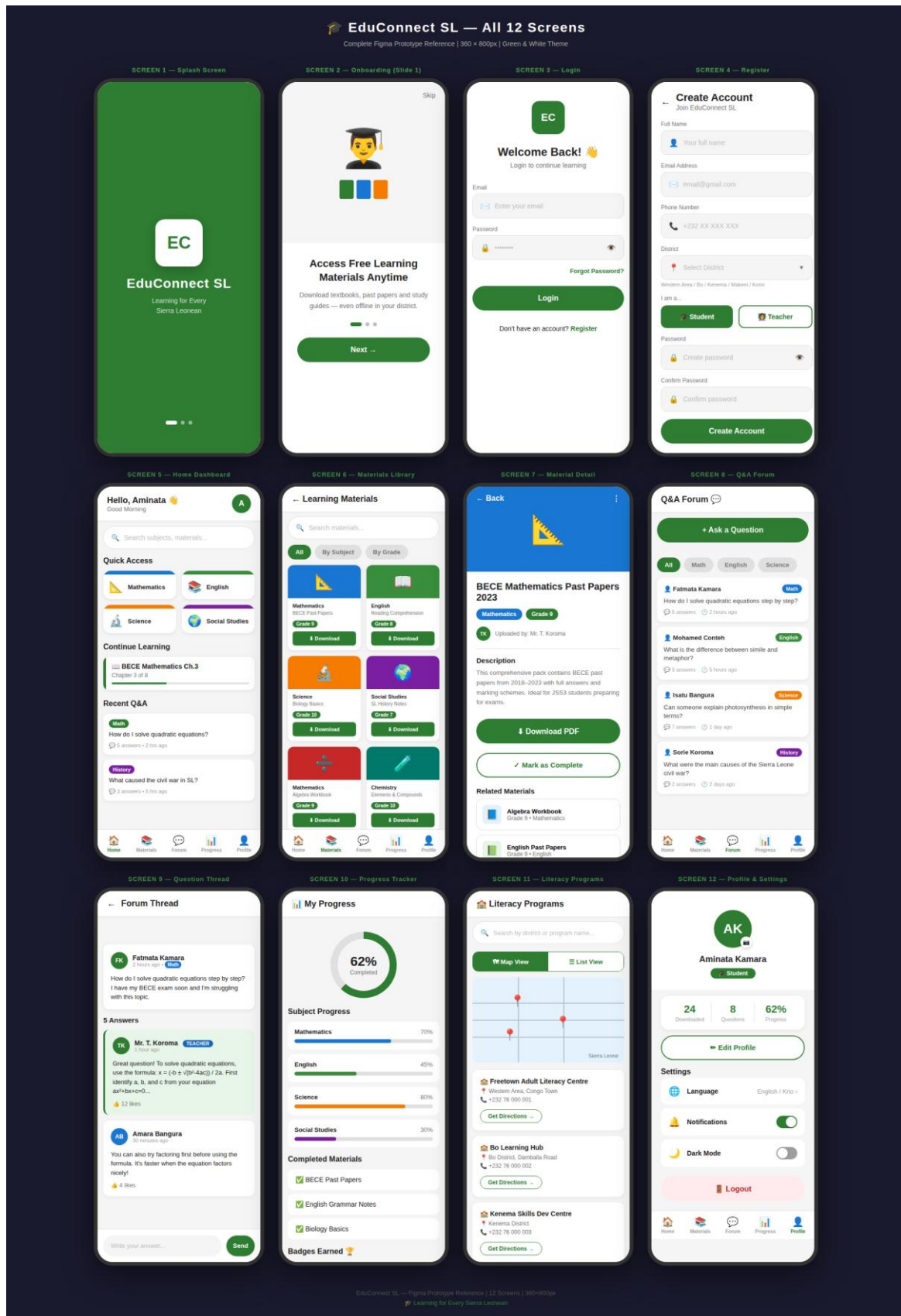
EduBridge was prototyped in two responsive formats to demonstrate cross-device usability, a critical requirement given that target users in Sierra Leone access digital platforms through both budget Android smartphones and shared desktop computers in school computer labs or internet cafés. The mobile version (500×950px) optimises for one-handed use, larger touch targets, and bottom navigation, while the desktop version takes advantage of wider screens to display additional contextual information, side navigation, and multi-column layouts. Both versions maintain identical branding, colour palette, and navigation logic to ensure a consistent user experience across devices.

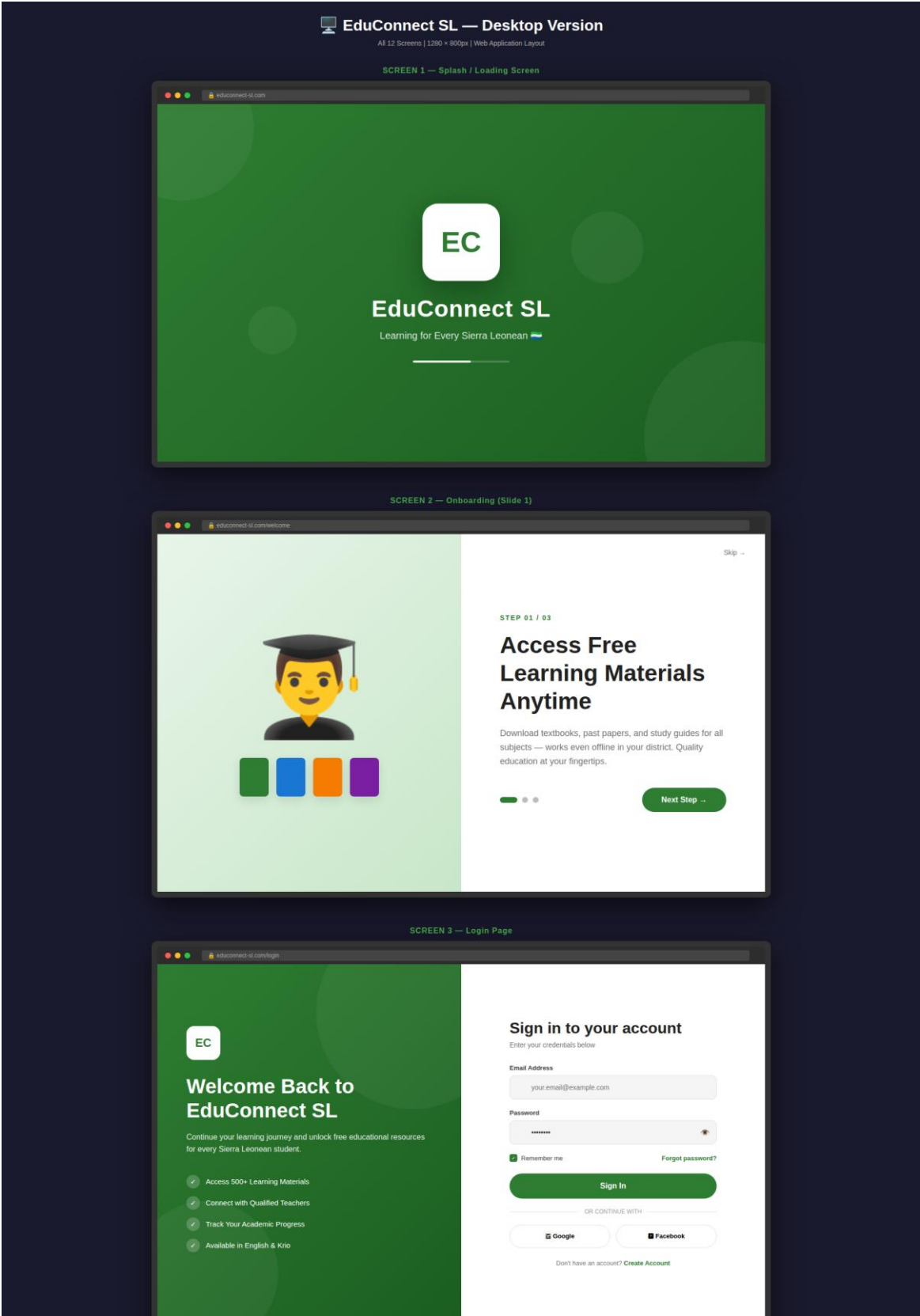
5.3 Figma Prototype Screens

The EduBridge prototype consists of 12 key screens covering all primary user journeys, designed in both mobile and desktop formats. The screens are organised into the following user flows: Onboarding (Screens 1–2), Authentication (Screens 3–4), Core Learning (Screens 5–8), Assessment (Screens 9–10), and Community Features (Screens 11–12).

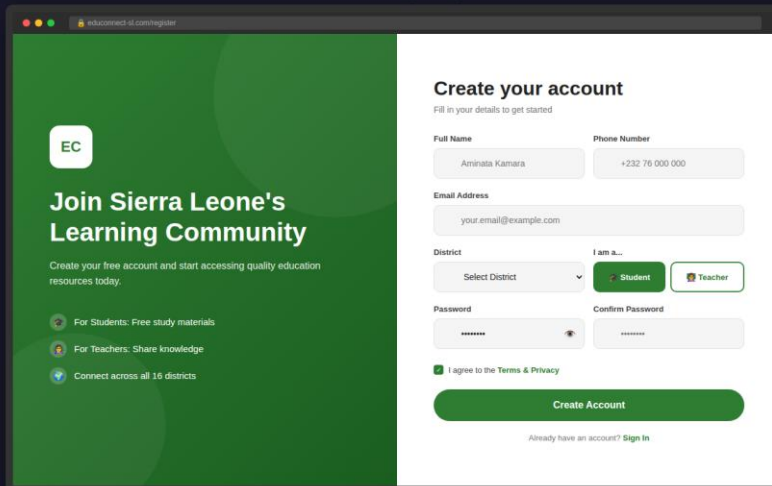
#	Screen Name	Purpose
1	Splash Screen	App branding and loading indicator on startup
2	Welcome / Landing Page	Introduce EduBridge with Get Started and Login options
3	Login Screen	Email and password authentication with Google OAuth option
4	Registration Screen	New user sign-up with role selection (Student or Mentor)
5	Student Dashboard	Overview of enrolled courses, progress stats, and upcoming quizzes
6	Course Catalog	Browse and filter courses by subject and grade level
7	Course Detail Page	Course description, lesson list, enrolment, and progress tracking

8	Learning Material Page	Video player, lesson notes, and downloadable resources
9	Quiz Page	Timed multiple-choice questions with progress indicator
10	Quiz Results Page	Score display, answer review, XP earned, and streak tracking
11	Mentor Matching Page	Browse available mentors by subject with request functionality
12	Community Forum	Discussion posts organised by subject with reply functionality





SCREEN 4 — Register Page



Create your account
Fill in your details to get started

Full Name: Aminata Kamara Phone Number: +232 76 000 000

Email Address: your.email@example.com

District: Select District I am a... ☒ Student ☐ Teacher

Password: Confirm Password

☒ I agree to the Terms & Privacy

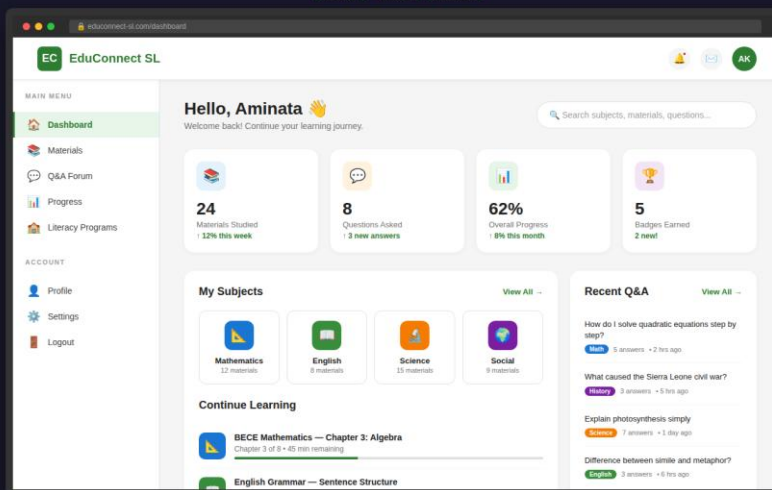
Create Account

Already have an account? [Sign In](#)

Join Sierra Leone's Learning Community
Create your free account and start accessing quality education resources today.

- For Students: Free study materials
- For Teachers: Share knowledge
- Connect across all 16 districts

SCREEN 5 — Student Dashboard



EduConnect SL Hello, Aminata 🙌
Welcome back! Continue your learning journey.

Search subjects, materials, questions...

24 Materials Studied
132% this week

8 Questions Asked
3 new answers

62% Overall Progress
8% this month

5 Badges Earned
2 new

My Subjects [View All](#)

- Mathematics (12 materials)
- English (8 materials)
- Science (11 materials)
- Social (9 materials)

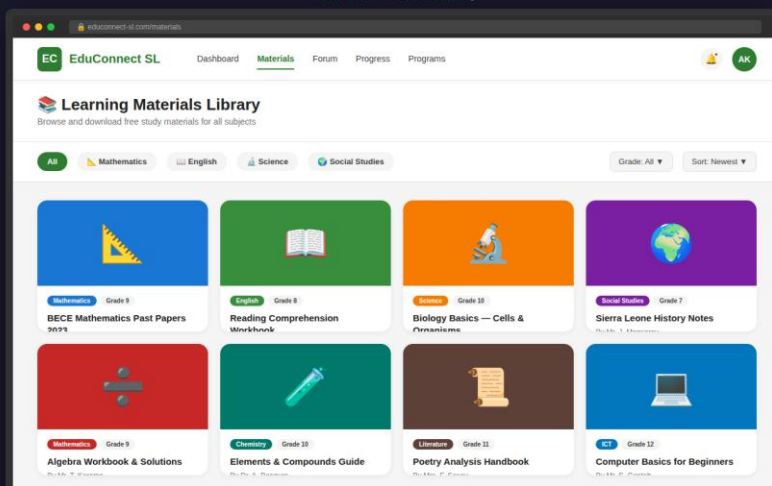
Continue Learning

- BECE Mathematics — Chapter 3: Algebra
Chapter 3 of 8 • 45 min remaining
- English Grammar — Sentence Structure

Recent Q&A [View All](#)

- How do I solve quadratic equations step by step?
[Math](#) 3 answers • 2 hrs ago
- What caused the Sierra Leone civil war?
[History](#) 3 answers • 5 hrs ago
- Explain photosynthesis simply
[Science](#) 7 answers • 1 day ago
- Difference between simile and metaphor?
[English](#) 3 answers • 6 hrs ago

SCREEN 6 — Materials Library



EduConnect SL Dashboard **Materials** Forum Progress Programs

Learning Materials Library
Browse and download free study materials for all subjects

All Mathematics English Science Social Studies

Grade: All Sort: Newest

- Mathematics** Grade 9: BECE Mathematics Past Papers (9979)
- English** Grade 8: Reading Comprehension Workbook
- Science** Grade 10: Biology Basics — Cells & Pyramids
- Social Studies** Grade 7: Sierra Leone History Notes
- Mathematics** Grade 9: Algebra Workbook & Solutions
- Chemistry** Grade 10: Elements & Compounds Guide
- Literature** Grade 11: Poetry Analysis Handbook
- ICT** Grade 12: Computer Basics for Beginners

SCREEN 7 — Material Detail Page

The screenshot shows the 'Materials' section of the EduConnect SL website. The main content area features a large blue header with a yellow ruler icon. Below this, the title 'BECE Mathematics Past Papers 2023 — Complete Solutions' is displayed, along with the author 'Mt. T. Koroma' and a 'Verified Teacher' badge. The page includes a 'Description' section and a 'Related Materials' sidebar on the right. The sidebar lists 'BECE Math 2022 Papers', 'Algebra Workbook', and 'BECE English Papers'. A 'Resource Access' section on the right indicates the material is 'FREE' and provides a 'Download PDF (12.5 MB)' button and a 'Mark as Complete' button.

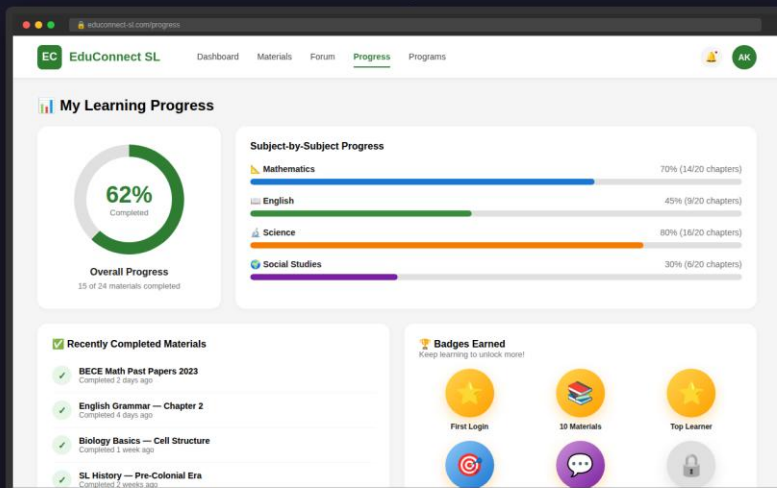
SCREEN 8 — Q&A Forum

The screenshot displays the 'Q&A Forum' section of the EduConnect SL website. The forum is categorized by subject: All Questions, Math, English, Science, Social Studies, and My Questions. A list of questions is shown, each with a user profile, a question title, a brief description, and the number of answers. The questions include topics like solving quadratic equations, simile and metaphor, photosynthesis, and the Sierra Leone civil war. A 'Top Contributors' sidebar on the right lists users like Mr. T. Koroma, Mrs. F. Sesay, and Dr. A. Bangura. A 'Forum Stats' section at the bottom right provides summary statistics: Total Questions: 1,234, Total Answers: 4,567, Active Users: 892, and Verified Teachers: 45.

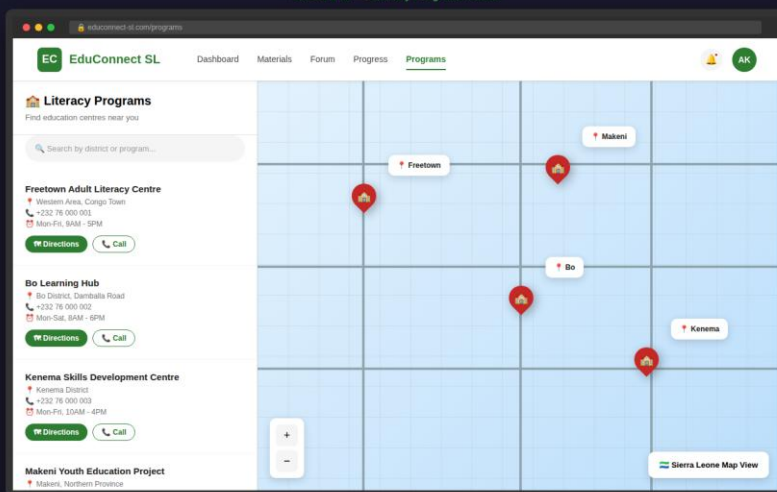
SCREEN 9 — Question Thread

The screenshot shows a specific question thread within the 'Math' category of the EduConnect SL forum. The question is titled 'How do I solve quadratic equations step by step?' and is posted by user 'Fatmata Kamara'. The thread shows 5 answers, with the first answer from 'Mr. T. Koroma' providing a detailed explanation of the quadratic formula. The answer includes the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and a step-by-step guide: Step 1: Identify a, b, and c from your equation; Step 2: Calculate the discriminant ($b^2 - 4ac$); Step 3: Substitute into the formula; Step 4: Solve for both values of x.

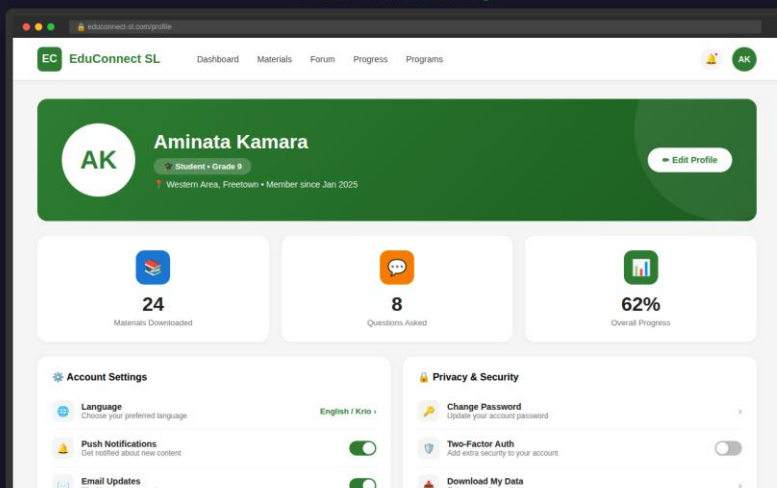
SCREEN 10 — Progress Tracker



SCREEN 11 — Literacy Programs Finder



SCREEN 12 — User Profile & Settings



5.4 Prototype Navigation Flow

All 12 screens are linked using Figma's Prototype feature to create a fully interactive, clickable flow. The Splash Screen is set as the starting frame. From the Welcome Screen, users can navigate to either Login or Registration; both paths lead to the Student Dashboard. From the Dashboard, the bottom navigation bar provides access to the Course Catalog, Mentor Matching, Community Forum, and Profile sections. The Course Catalog leads to Course Detail, which leads to the Learning Material page and then the Quiz Page, which finally leads to the Quiz Results page. This flow was replicated identically across both the mobile and desktop prototypes.

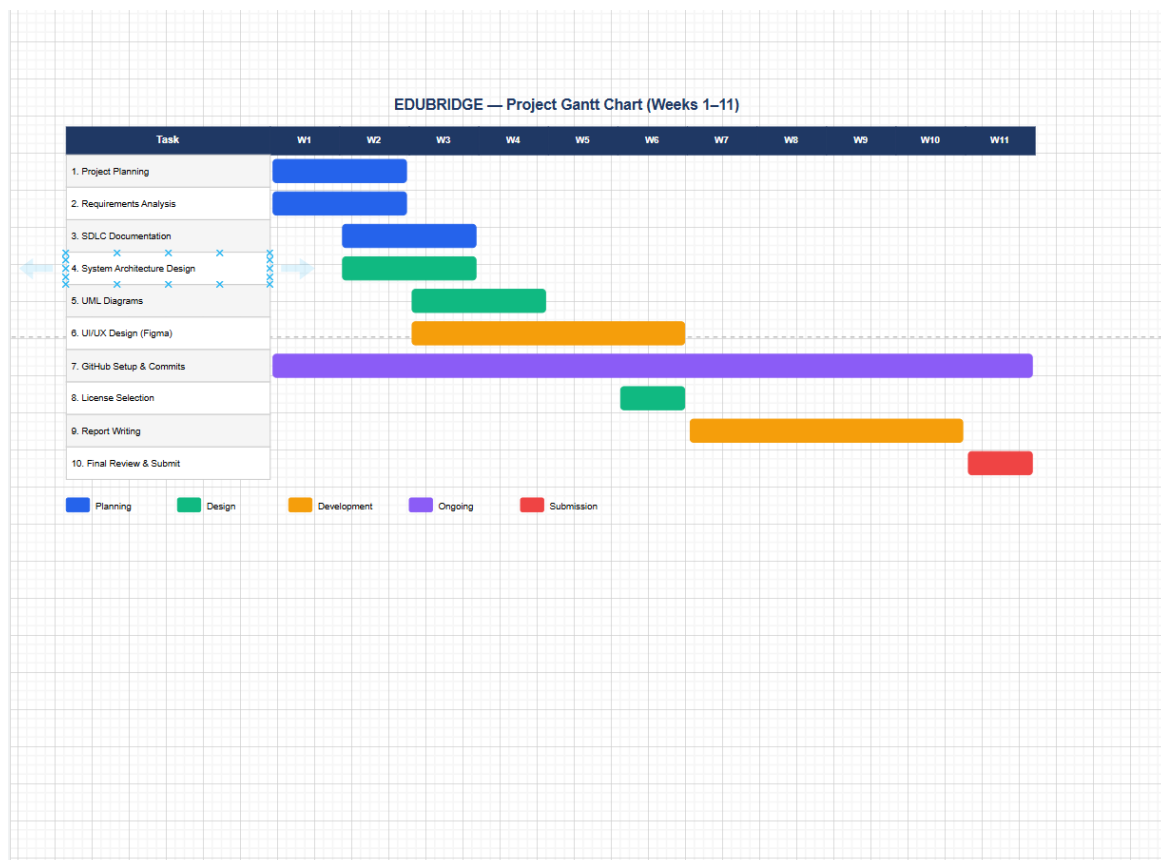
Mobile	Prototype	Link:
		[https://www.figma.com/proto/Cvyc8nHdXaAZCkk4dqXVGR/EduBridge-%E2%80%93BBIT1201-Project?page-id=0%3A1&node-id=1-2&viewport=582%2C233%2C0.54&t=dFe7zLKnd79cpRgy-1&scaling=scale-down&content-scaling=fixed]

Desktop	Prototype	Link:
		[https://www.figma.com/design/Cvyc8nHdXaAZCkk4dqXVGR/EduBridge-%E2%80%93BBIT1201-Project?node-id=213-857&t=TrLsH7a5EyeJb4qf-1]

6. Project Management

6.1 Gantt Chart

The Gantt chart below illustrates the planned timeline for the EduBridge project across eleven weeks. Tasks are grouped into four categories: planning activities, development activities, testing and deployment, and ongoing activities. GitHub integration and project documentation run continuously throughout the entire project duration, reflecting their importance at every stage.

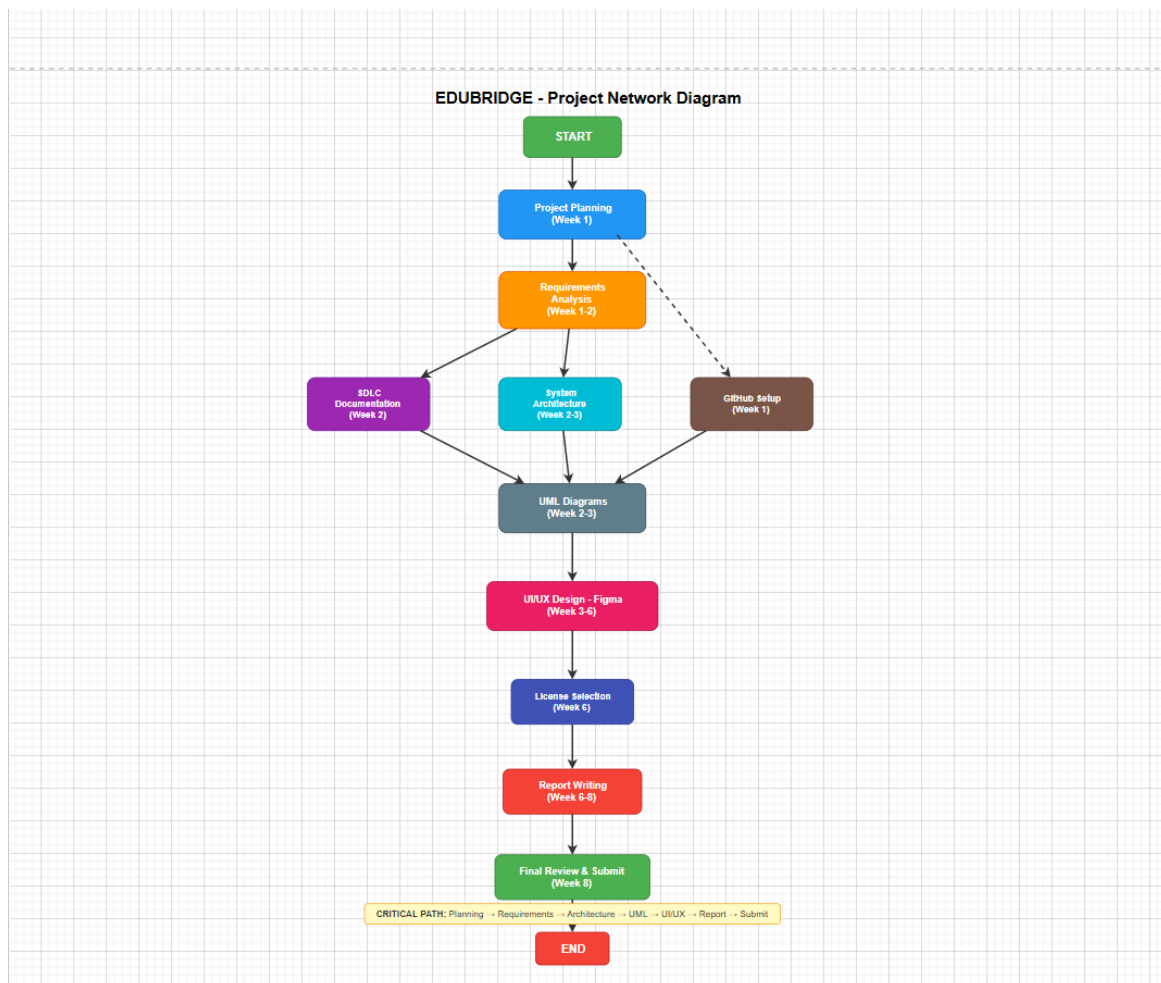


6.2 Network Diagram

The network diagram maps the dependencies between all project tasks using the Critical Path Method (CPM). The critical path runs through: Project Planning → Requirements Analysis → System Architecture Design → UML Diagrams → UI/UX Design (Figma) → License Selection → Report Writing → Final Review and

Submission. Any delay along this path will directly impact the Week 11 submission deadline.

Parallel, non-critical activities such as GitHub Setup and SDLC Documentation can experience minor delays without affecting the final deadline, provided they are completed before the Report Writing phase begins.



7. GitHub and Version Control

7.1 Repository Structure

EduBridge is hosted in a public GitHub repository with the following organised folder structure, enabling clear navigation and demonstrating professional software project management:

Folder / File	Contents
README.md	Project overview, SDG alignment, features, prototype links, tech stack, backend structure
LICENSE	MIT License — open-source licence for Digital Public Goods compliance
docs/	EduBridge_Final_Report.pdf — the complete project report
diagrams/	Architecture, Use Case, Class, Sequence, and Activity diagram PNG files
prototype/	figma_link.txt — mobile and desktop Figma prototype share links
project-management/	Gantt chart and Network diagram image files
design/	Screenshots of all 12 mobile and desktop Figma prototype screens
backend/	Node.js + Express server, routes, and PostgreSQL database schema

Repository Link: [[Link](#)]

7.2 Backend Folder Structure

To demonstrate a complete, buildable system architecture, the GitHub repository includes a dedicated `/backend` folder containing the Node.js and Express server structure, all six API route modules, and the full PostgreSQL database schema. This allows the marker to verify that the system design extends beyond the user interface into a fully specified, production-ready backend.

```
backend/
├── server.js      (Main API server entry point)
├── package.json   (Project dependencies)
├── routes/
│   ├── auth.js    (Register and login)
│   ├── courses.js (Browse, view, enroll)
│   ├── quiz.js    (Load quiz, submit answers)
│   ├── mentor.js  (Browse mentors, send request)
│   └── forum.js   (View posts, create post)
├── database/
│   └── schema.sql (9 tables + sample data)
└── API_TO_SCREEN_MAP.md (Maps Figma screens to API endpoints)
```

The database schema defines nine PostgreSQL tables — Users, Courses, Enrollments, Materials, Quizzes, Quiz_Results, Progress, Mentorship, and Forum_Posts — connected through foreign key relationships. For example, the Enrollments table references both the Users table and the Courses table, establishing the link between a student and their enrolled courses. Each backend route file documents the intended SQL queries as inline comments, for example `SELECT * FROM courses WHERE is_published = true` for retrieving the course catalog, demonstrating the planned interaction between the Application Layer and the Data Layer.

7.3 Mapping Figma Screens to Backend Endpoints

Because the Figma prototype and the backend code exist as two separate artefacts, a dedicated mapping document (backend/API_TO_SCREEN_MAP.md) was created to explicitly show which backend endpoint each prototype screen would call in a fully deployed version of the system. This demonstrates a clear understanding of full-stack architecture without requiring functional code integration.

Figma Screen	User Action	Backend Endpoint
Login Screen	User clicks Login	POST /api/auth/login
Registration Screen	User clicks Register	POST /api/auth/register
Student Dashboard	Page loads	GET /api/courses (enrolled)
Course Catalog	Page loads	GET /api/courses
Course Detail Page	User clicks Enroll	POST /api/courses/:id/enroll
Learning Material Page	User clicks Download	GET /api/materials/:id
Quiz Page	Page loads / Submit	GET /api/quiz/:id, POST /api/quiz/submit
Mentor Matching Page	Page loads / Request	GET /api/mentor, POST /api/mentor/request
Community Forum	Page loads / Post	GET /api/forum/posts, POST /api/forum/post

7.4 Commit History

The repository demonstrates a progressive and structured commit history showing active collaboration and incremental development across all project phases:

Week	Commit Message
Week 1	Initial project setup — README and folder structure created
Week 1	Add requirements analysis and stakeholder register
Week 2	Add system architecture diagram and component descriptions
Week 2	Add SDLC methodology — Agile model documentation
Week 3	Add Use Case diagram and Class diagram
Week 3	Add Sequence diagram — student takes a quiz
Week 4	Begin Figma prototype — mobile splash, welcome, and login screens
Week 5	Add desktop versions of all screens for responsive design
Week 5	Add registration, dashboard, course catalog, and course detail screens
Week 6	Add learning material, quiz, results, mentor, and forum screens
Week 6	Add MIT License file

Week 7	Add main server file for EduBridge backend
Week 7	Add package.json with backend dependencies
Week 7	Add authentication, course, quiz, mentor, and forum routes
Week 8	Add PostgreSQL database schema with all 9 tables and sample data
Week 8	Add Figma screen to backend API mapping document
Week 9	Add Gantt chart and Network diagram to project-management folder
Week 10	Complete final report draft and upload to docs folder
Week 11	Final submission — all deliverables uploaded and verified

8. License, Design Quality, and SDG Relevance

8.1 Software License

EduBridge is released under the MIT License. This licence was selected because it is one of the most permissive and widely recognised open-source licences in the software industry. It permits anyone to freely use, copy, modify, merge, publish, distribute, sublicense, and sell copies of the software, provided that the original copyright notice and permission notice are included in all copies or substantial portions of the software.

This choice aligns directly with the Digital Public Goods Alliance (DPGA) standard, which requires that all recognised digital public goods use an approved open-source licence. The MIT Licence ensures that governments, NGOs, schools, and individual developers across Sierra Leone and the broader West African region can adopt, adapt, and redistribute EduBridge at no cost, maximising its public benefit and reach.

All third-party resources used in this project are properly attributed: icons sourced from Flaticon (Flaticon Free Licence), fonts from Google Fonts (SIL Open Font Licence), and placeholder images sourced from Unsplash (Unsplash Licence — free for both commercial and non-commercial use without attribution required).

8.2 Design Quality (UI/UX)

The EduBridge user interface was designed with three guiding principles: simplicity, visual consistency, and cross-device accessibility. Every design decision prioritises the experience of a first-time user in Sierra Leone, whether on a budget Android smartphone or a shared desktop computer.

The colour palette uses Primary Blue (#2563EB) as the brand colour, chosen for its association with education, trust, and technology. Secondary Green (#10B981) signals progress and positive outcomes. All colour combinations comply with WCAG 2.1 AA contrast ratio standards, ensuring accessibility for users with visual impairments.

Navigation follows a bottom navigation bar pattern on mobile and an adapted side or top navigation pattern on desktop, placing the five core sections — Home, Courses, Mentor, Forum, and Profile — within easy reach on every device. All button and input

elements maintain a minimum tap target height of 48px to prevent accidental presses on small screens. Card-based layouts provide clear visual boundaries between content items and remain consistent across all 12 screens in both responsive formats, creating a unified and professional visual language.

8.3 SDG 4 — Quality Education Alignment

EduBridge directly addresses United Nations Sustainable Development Goal 4: Quality Education. The SDG 4 mandate calls for inclusive and equitable quality education and promotion of lifelong learning opportunities for all. In Sierra Leone, significant challenges persist: the net enrolment rate in secondary education remains below 50%, many rural communities lack qualified teachers, and physical textbooks are often unavailable or unaffordable.

Target 4.1 — Free and Equitable Primary and Secondary Education:

EduBridge provides a completely free library of curriculum-aligned learning materials organised by subject and grade level. Students in the most remote districts of Sierra Leone can access the same quality resources as students in Freetown, directly eliminating the geographic inequality that currently defines educational outcomes in the country.

Target 4.3 — Equal Access to Affordable Technical and Vocational Education: The platform's mentor matching system and ICT-focused course offerings allow students to develop practical, employment-relevant skills including programming, digital literacy, and data management, preparing them for the modern economy.

Target 4.5 — Eliminate Gender Disparities and Barriers for Vulnerable Populations: EduBridge is designed to be accessible to all students regardless of gender, geographic location, or socioeconomic status. The platform is free, available on both mobile and desktop, and optimised for low-bandwidth connections, removing the primary barriers that disproportionately affect girls and rural students in Sierra Leone.

9. Conclusion

9.1 Summary

EduBridge is a comprehensive Digital Public Good designed to address the education crisis facing students in Sierra Leone and similar underserved communities worldwide. This report has presented a complete software design covering the Agile SDLC methodology, a scalable 3-tier system architecture, four UML diagrams, a responsive 12-screen Figma prototype built for both mobile and desktop, a structured GitHub repository with a fully documented backend folder and database schema, a Gantt chart and network diagram for project planning, and full documentation of the MIT licence, UI/UX design decisions, and SDG 4 alignment.

The design demonstrates that a free, responsive, low-bandwidth digital learning platform is technically feasible and can be built using entirely open-source tools. By addressing Targets 4.1, 4.3, and 4.5 of SDG 4, EduBridge has the potential to meaningfully improve educational outcomes for thousands of students across Sierra Leone.

9.2 Challenges

- Designing for low-bandwidth connectivity required careful consideration of asset sizes, caching strategies, and progressive loading patterns.
- Balancing the depth of features with the simplicity required for first-time users in low-digital-literacy environments was a constant design tension.
- Maintaining identical navigation logic and visual consistency across both mobile and desktop prototype versions required disciplined design system management.
- Documenting the relationship between the Figma prototype and the backend API design, without building functional integration, required careful and explicit mapping.

9.3 Future Work

- Implementation of a fully functional offline mode using Service Workers and local storage caching.
- Multilingual support in Krio, Temne, and Mende to serve Sierra Leone's diverse linguistic communities.
- Gamification features including XP points, achievement badges, and weekly leaderboards to increase student motivation.
- Integration with the Ministry of Basic and Senior Secondary Education (MBSSE) curriculum database for automatically aligned content.
- Full implementation of the backend API and connection to the live database, moving from design to deployment.
- Submission of EduBridge to the Digital Public Goods Alliance for official DPG recognition.

9.4 References

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